
STM32
CubeMX

1. Description

1.1. Project

Project Name	robot_mcu
Board Name	custom
Generated with:	STM32CubeMX 6.18.1
Date	10/18/2026

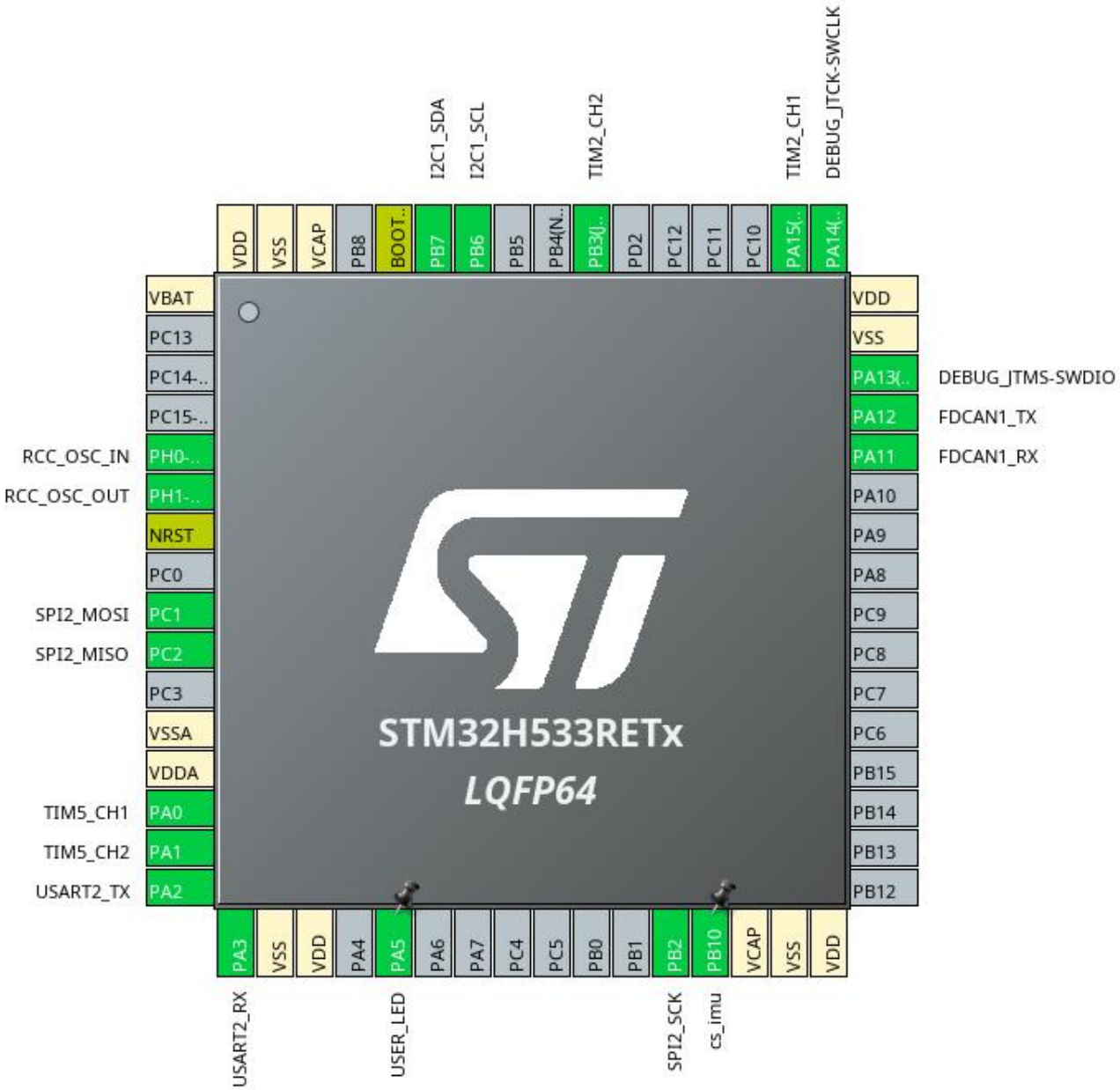
1.2. MCU

MCU Series	STM32H5
MCU Line	STM32H5x3
MCU name	STM32H533RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M33

2. Pinout Configuration



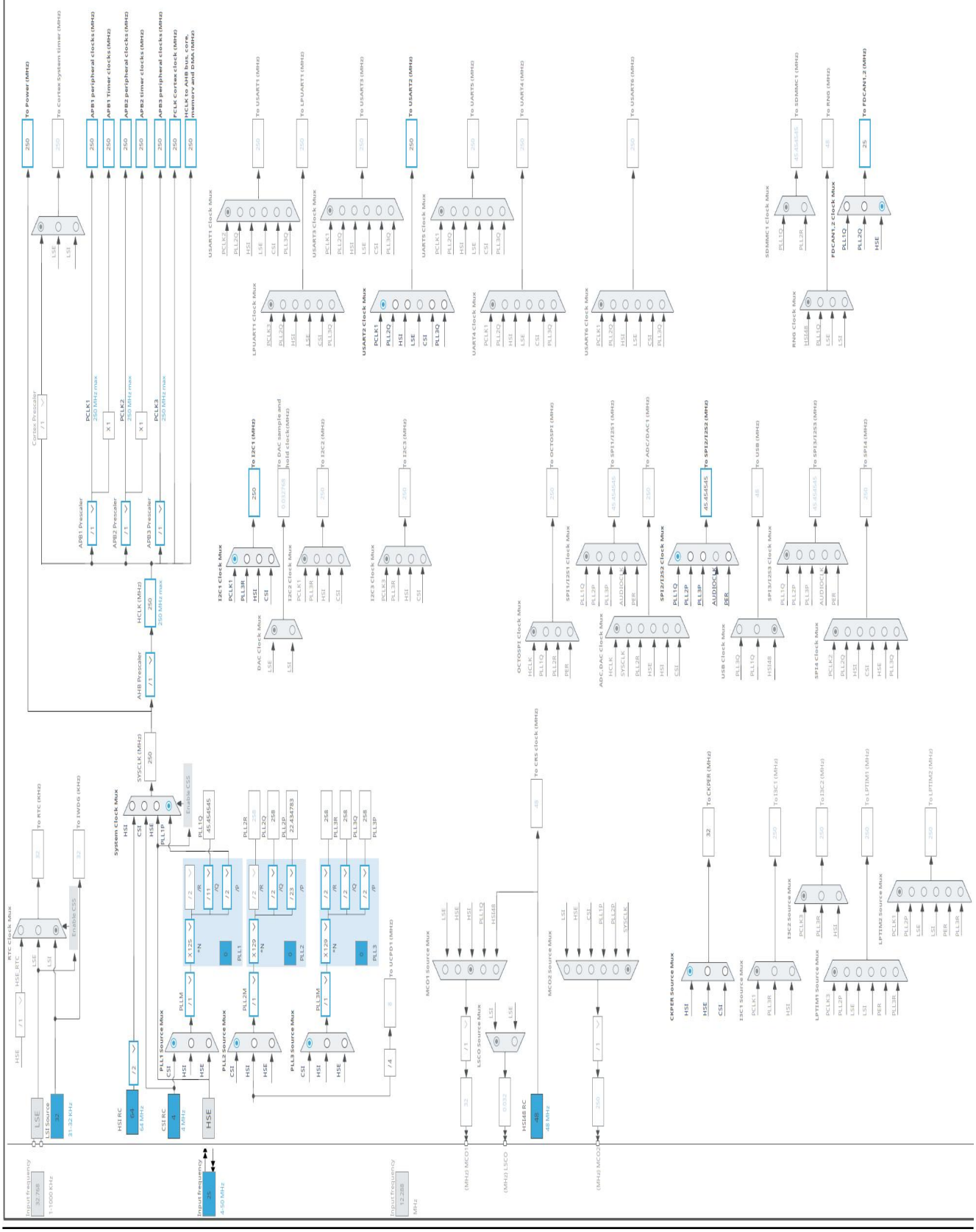
3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
5	PH0-OSC_IN(PH0)	I/O	RCC_OSC_IN	
6	PH1-OSC_OUT(PH1)	I/O	RCC_OSC_OUT	
7	NRST	Reset		
9	PC1	I/O	SPI2_MOSI	
10	PC2	I/O	SPI2_MISO	
12	VSSA	Power		
13	VDDA	Power		
14	PA0	I/O	TIM5_CH1	
15	PA1	I/O	TIM5_CH2	
16	PA2	I/O	USART2_TX	
17	PA3	I/O	USART2_RX	
18	VSS	Power		
19	VDD	Power		
21	PA5 *	I/O	GPIO_Output	USER_LED
28	PB2	I/O	SPI2_SCK	
29	PB10 *	I/O	GPIO_Output	cs_imu
30	VCAP	Power		
31	VSS	Power		
32	VDD	Power		
44	PA11	I/O	FDCAN1_RX	
45	PA12	I/O	FDCAN1_TX	
46	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
47	VSS	Power		
48	VDD	Power		
49	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
50	PA15(JTDI)	I/O	TIM2_CH1	
55	PB3(JTDO/TRACESWO)	I/O	TIM2_CH2	
58	PB6	I/O	I2C1_SCL	
59	PB7	I/O	I2C1_SDA	
60	BOOT0	Boot		
62	VCAP	Power		
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32H533RETx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H5
Line	STM32H5x3
MCU	STM32H533RETx
Datasheet	DS00000_Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

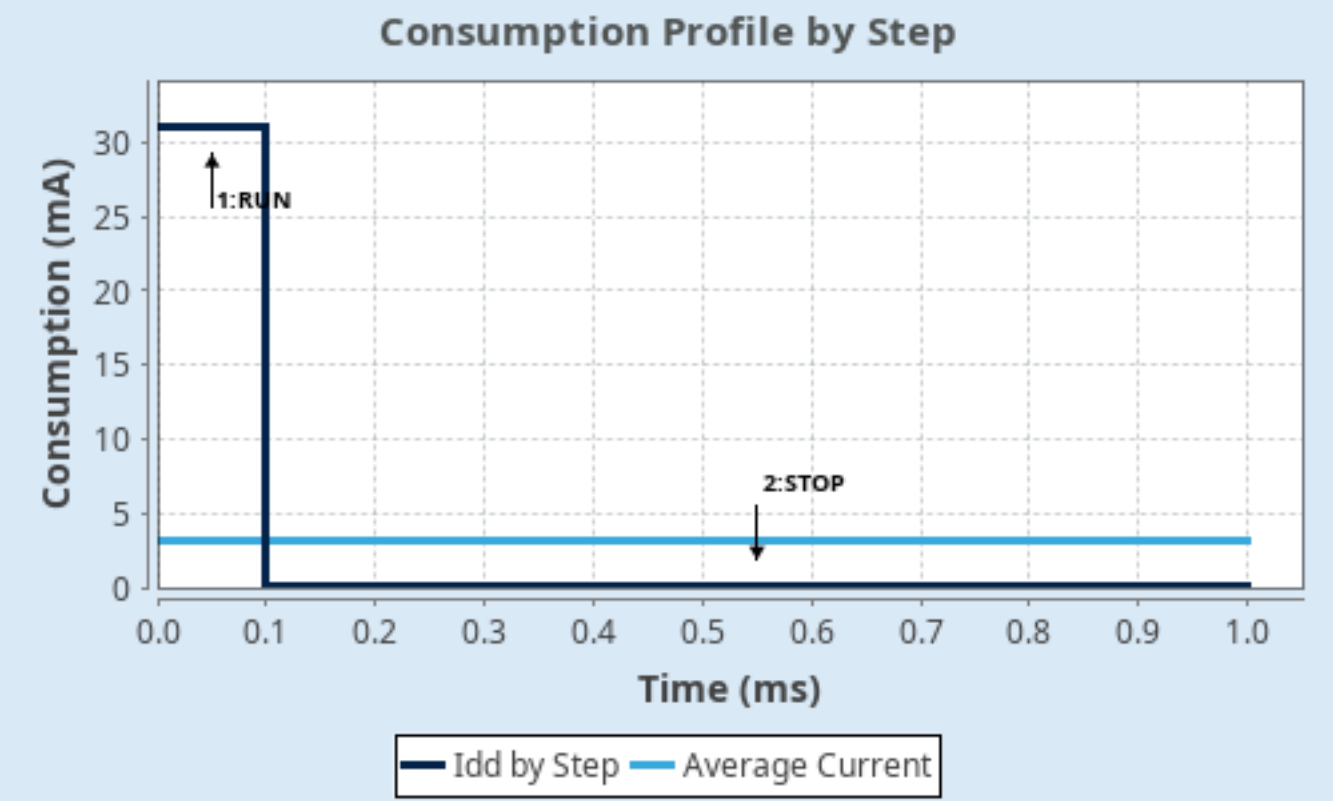
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
Fetch Type	FLASH_ON/Cache2Ways_ALL_RAM_RETENTION	FLASH_PwrDwnStop_OFF
CPU Frequency	250 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	31 mA	100 µA
Duration	0.1 ms	0.9 ms
DMIPS	375.0	0.0
Ta Max	125	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.19 mA
Battery Life	1 month, 13 days, 22 hours	Average DMIPS	37.5 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
30
25
20
15
10
5
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	robot_mcu
Project Folder	/home/bastien/Documents/robot_mcu
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H5 V1.7.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x1000
Minimum Stack Size	0x1000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_GPDMA1_Init	GPDMA1
4	MX_ICACHE_Init	ICACHE
5	MX_USART2_UART_Init	USART2
6	MX_TIM2_Init	TIM2
7	MX_TIM5_Init	TIM5
8	MX_SPI2_Init	SPI2
9	MX_FDCAN1_Init	FDCAN1
10	MX_I2C1_Init	I2C1
11	MX_TIM6_Init	TIM6

3. Peripherals and Middlewares Configuration

3.1. BOOTPATH

mode: Activated

3.2. CORTEX_M33_NS

SysTick Source Selection: HCLK

3.2.1. Parameter Settings:

OTP/RO areas default mode Settings:

Enable OTP/RO area cache disable	**Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

Region	Enabled
Region Base Address	**0x08FFF000 ***
Region Limit Address	**0x08FFFFFF ***
Attributes Number	ATTRIBUTE 0
Access Permission	ALL READS Permissions
Instruction Access	Disable
Shareability Permission	Disable

Cortex Memory Protection Unit Region 1 Settings:

Region	Disabled

Cortex Memory Protection Unit Region 2 Settings:

Region	Disabled

Cortex Memory Protection Unit Region 3 Settings:

Region	Disabled

Cortex Memory Protection Unit Region 4 Settings:

Region	Disabled

Cortex Memory Protection Unit Region 5 Settings:

Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

Region	Disabled

3.2.2. Attributes Settings:

Cortex Memory Protection Unit Attributes 0 Settings:

Attributes	Enabled
Memory type	Normal non-cacheable

Cortex Memory Protection Unit Attributes 1 Settings:

Attributes	Disabled

Cortex Memory Protection Unit Attributes 2 Settings:

Attributes	Disabled

Cortex Memory Protection Unit Attributes 3 Settings:

Attributes	Disabled

Cortex Memory Protection Unit Attributes 4 Settings:

Attributes	Disabled

Cortex Memory Protection Unit Attributes 5 Settings:

Attributes	Disabled

Cortex Memory Protection Unit Attributes 6 Settings:

Attributes	Disabled

Cortex Memory Protection Unit Attributes 7 Settings:

Attributes	Disabled

3.3. DEBUG

Debug: Serial Wire

3.4. FDCAN1

mode: Activated

3.4.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1

Std Filters Nbr	0
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode

Bit Timings Parameters:

Nominal Prescaler	**8 ***
Nominal Time Quantum	**320.0 ***
Nominal Time Seg1	1
Nominal Time Seg2	1
Nominal Time for one Bit	**960 ***
Nominal Baud Rate	**1041666 ***

3.5. I2C1

I2C: I2C

3.5.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x60808CD3 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.6. ICACHE

Mode: 1-way (direct mapped cache)

3.7. MEMORYMAP

mode: Activated

3.8. PWR

mode: Power saving mode

mode: Privilege attributes

3.8.1. PWR Privilege :

Privilege PWR:

PWR Privilege	Disable

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.9.1. RCC Privilege :

Privilege RCC:

Privilege of RCC Non-Secure Items	Disable

3.9.2. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	5 WS (6 CPU cycle)
Flash Programming Delay	2

RCC Parameters:

HSI Calibration Value	64
CSI Calibration Value	32
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
TIM Prescaler Selection	Disabled

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

PLL1/2/3 Parameters:

PLL1 input frequency range	Between 4 and 8 MHz

3.10. SPI2

Mode: Full-Duplex Master

3.10.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	**8 Bits ***
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	**22.727272 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

CRC Parameters:

CRC Calculation	Disabled

Advanced Parameters:

NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.11. SYS

Timebase Source: TIM1

3.12. TIM2

Combined Channels: Encoder Mode

3.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295

Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	**Encoder Mode TI1 and TI2 ***
Slave Mode Preload Activation	Disable
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.13. TIM5

Combined Channels: Encoder Mode

3.13.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	**Encoder Mode TI1 and TI2 ***
Slave Mode Preload Activation	Disable
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct

Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.14. TIM6

mode: Activated

3.14.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**249 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	**999 ***
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)

3.15. USART2

Mode: Asynchronous

3.15.1. Parameter Settings:

Basic Parameters:

Baud Rate	**2000000 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration

Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.16. STMicroelectronics.X-CUBE-FREERTOS.1.6.0
mode: CMSISJjRTOS2

3.16.1. Config parameters:

API:

FreeRTOS API	CMSIS-RTOS2

Versions:

FreeRTOS version	11.2.0
CMSIS-RTOS version	2.1.3

Security settings:

RUN_FREERTOS_SECURE_ONLY	Disabled

MPU/FPU:

ENABLE_FPU	Disabled

MVE:

configENABLE_MVE	Disabled

Kernel settings:

MINIMAL_SECURE_STACK_SIZE	**128 ***
USE_PREEMPTION	Enabled
USE_TIME_SLICING	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	**0x1000 ***
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled

USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
USE_MINI_LIST_ITEM	Enabled
USE_SB_COMPLETED_CALLBACK	Disabled
USE_EVENT_GROUPS	Enabled
USE_STREAM_BUFFERS	Enabled
KERNEL_PROVIDED_STATIC_MEMORY	Enabled
STATS_BUFFER_MAX_LENGTH	**0xFFFF ***
CHECK_HANDLER_INSTALLATION	Enabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	**100000 ***
STACK_ALLOCATION_FROM_SEPARATE_HEAP	Disabled
HEAP_CLEAR_MEMORY_ON_FREE	Disabled
ENABLE_HEAP_PROTECTOR	Disabled
VALIDATE_HEAP_BLOCK_POINTER	Disabled

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled
RUN_TIME_COUNTER_TYPE	uint32_t

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	**2 ***
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	4096

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.16.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
xTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Enabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Enabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.16.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings (see parameter description first):

Use pack heap file	Enabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DEBUG	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
FDCAN1	PA11	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA12	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
RCC	PH0-OSC_IN(PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT(PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SPI2	PC1	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC2	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB2	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM2	PA15(JTDI)	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB3(JTDO/TRACESWO)	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM5	PA0	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1	TIM5_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USER_LED
	PB10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	cs_imu

4.2. GPDMA1

Channel 3 - 2 Words Internal FIFO : Standard Request Mode
Channel 2 - 2 Words Internal FIFO : Standard Request Mode
Channel 1 - 2 Words Internal FIFO : Standard Request Mode
Channel 0 - 2 Words Internal FIFO : Standard Request Mode

4.2.1. All Channels:

Channel 0:

Request	GPDMA1_REQUEST_USART2_TX

Channel 1:

Request	GPDMA1_REQUEST_USART2_RX

Channel 2:

Request	GPDMA1_REQUEST_SPI2_TX

Channel 3:

Request	GPDMA1_REQUEST_SPI2_RX

4.2.2. SECURITY:

CH3:

Enable Channel as Privileged	NON PRIVILEDGED

CH2:

Enable Channel as Privileged	NON PRIVILEDGED

CH1:

Enable Channel as Privileged	NON PRIVILEDGED

CH0:

Enable Channel as Privileged	NON PRIVILEDGED

4.2.3. CH3:

Circular configuration:

Circular Mode	Disable

Request Configuration:

Request	**SPI2_RX/I2S2_RX ***
DMA Handle in IP Structure	hdmarx
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	**Medium ***
Transaction Mode	Normal
Direction	Peripheral To Memory

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	**Enabled ***
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

4.2.4. CH2:

Circular configuration:

Circular Mode	Disable

Request Configuration:

Request	**SPI2_TX/I2S2_TX ***
DMA Handle in IP Structure	hdmatx
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	**Medium ***
Transaction Mode	Normal
Direction	**Memory To Peripheral ***

Source Data Setting:

Source Address Increment After Transfer	**Enabled ***
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

4.2.5. CH1:

Circular configuration:

Circular Mode	**Enable ***
Circular Port	Port 0

Request Configuration:

Request	**USART2_RX ***
DMA Handle in IP Structure	hdmarx
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	**Very High ***
Transaction Mode	Normal
Direction	Peripheral To Memory
Node Type	GPDMA Linear Addressing

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	**Enabled ***
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

Channel Configuration for Linked List:

Execution Mode (circular/linear) of the Linked List	Circular
Linked List Execution Mode	The List is fully executed

4.2.6. CH0:

Circular configuration:

Circular Mode	**Enable ***
Circular Port	Port 0

Request Configuration:

Request	**USART2_TX ***
DMA Handle in IP Structure	hdmatx
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	**Very High ***
Transaction Mode	Normal
Direction	**Memory To Peripheral ***
Node Type	GPDMA Linear Addressing

Source Data Setting:

Source Address Increment After Transfer	**Enabled ***
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

Channel Configuration for Linked List:

Execution Mode (circular/linear) of the Linked List	Circular
Linked List Execution Mode	The List is fully executed

4.3. GPDMA2

4.4. LINKEDLIST

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
GPDMA1 Channel 0 global interrupt	true	8	0
GPDMA1 Channel 1 global interrupt	true	8	0
GPDMA1 Channel 2 global interrupt	true	5	0
GPDMA1 Channel 3 global interrupt	true	5	0
FDCAN1 interrupt 0	true	5	0
FDCAN1 interrupt 1	true	5	0
TIM1 Update interrupt	true	15	0
TIM2 global interrupt	true	5	0
TIM5 global interrupt	true	5	0
TIM6 global interrupt	true	5	0
SPI2 global interrupt	true	5	0
USART2 global interrupt	true	9	0
Flash non-secure global interrupt	unused		
RCC non-secure global interrupt	unused		
I2C1 Event interrupt	unused		
I2C1 Error interrupt	unused		
FPU global interrupt	unused		
Instruction cache global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
GPDMA1 Channel 0 global interrupt	false	true	true
GPDMA1 Channel 1 global interrupt	false	true	true
GPDMA1 Channel 2 global interrupt	false	true	true
GPDMA1 Channel 3 global interrupt	false	true	true
FDCAN1 interrupt 0	false	true	true
FDCAN1 interrupt 1	false	true	true
TIM1 Update interrupt	false	true	true
TIM2 global interrupt	false	true	true
TIM5 global interrupt	false	true	true
TIM6 global interrupt	false	true	true
SPI2 global interrupt	false	true	true
USART2 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-FREERTOS	1.6.0	Class : CMSIS Group : RTOS2 SubGroup : Heap Variant : Heap_1 Version : 11.2.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h5-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h5.pdf
Security Advisory	https://www.st.com/resource/en/security_advisory/sa0055-sfi-security-enhancement-for-stm32h523533-microcontrollers-stm32u5-series-and-stm32l5-series-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-

	stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0067-applicability-of-threadx-middleware-vulnerabilities-on-stm32cube-packages-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5930-guidelines-for-

	power-management-on-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5711-getting-started-with-stm32h5-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5872-introduction-to-the-system-architecture-and-performance-in-the-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4894-how-to-use-eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6007-getting-started-with-stirot-st-immutable-root-of-trust-for-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4992-introduction-to-secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6014-introduction-to-sfsp-versions-for-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5593-how-to-use-the-gpdma-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5879-introduction-to-i3c-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5959-migrating-within-the-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5992-migrating-from-stm32f446-to-stm32h523533-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6008-getting-started-with-debug-authentication-da-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an6202-how-to-calibrate-internal-rc-oscillators-on-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6246-guidelines-for-estimating-stm32h5-mcus-lifetime-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-how-to-use-usart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-how-to-use-usb-dfu-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-introduction-to-debug-toolbox-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5050-getting-started-with-octospi-hexadecaspi-and-xspi-interfaces-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5212-how-to-use-the-

	cache-to-optimize-performance-and-power-efficiency-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-how-to-use-i3c-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6308-migrating-from-stm32h5-to-stm32c5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-mcus-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0621-stm32h523xx-and-stm32h533xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm01036966.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0264-stm32-cortexm33-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0481-stm32h52333xx-stm32h56263xx-and-stm32h573xx-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-

	stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-devices-in-bga-packages-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3150-stm32h5-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3267-stm32h5-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3299-stm32h533xx-security-guidance-for-sesip-3-certification-stmicroelectronics.pdf